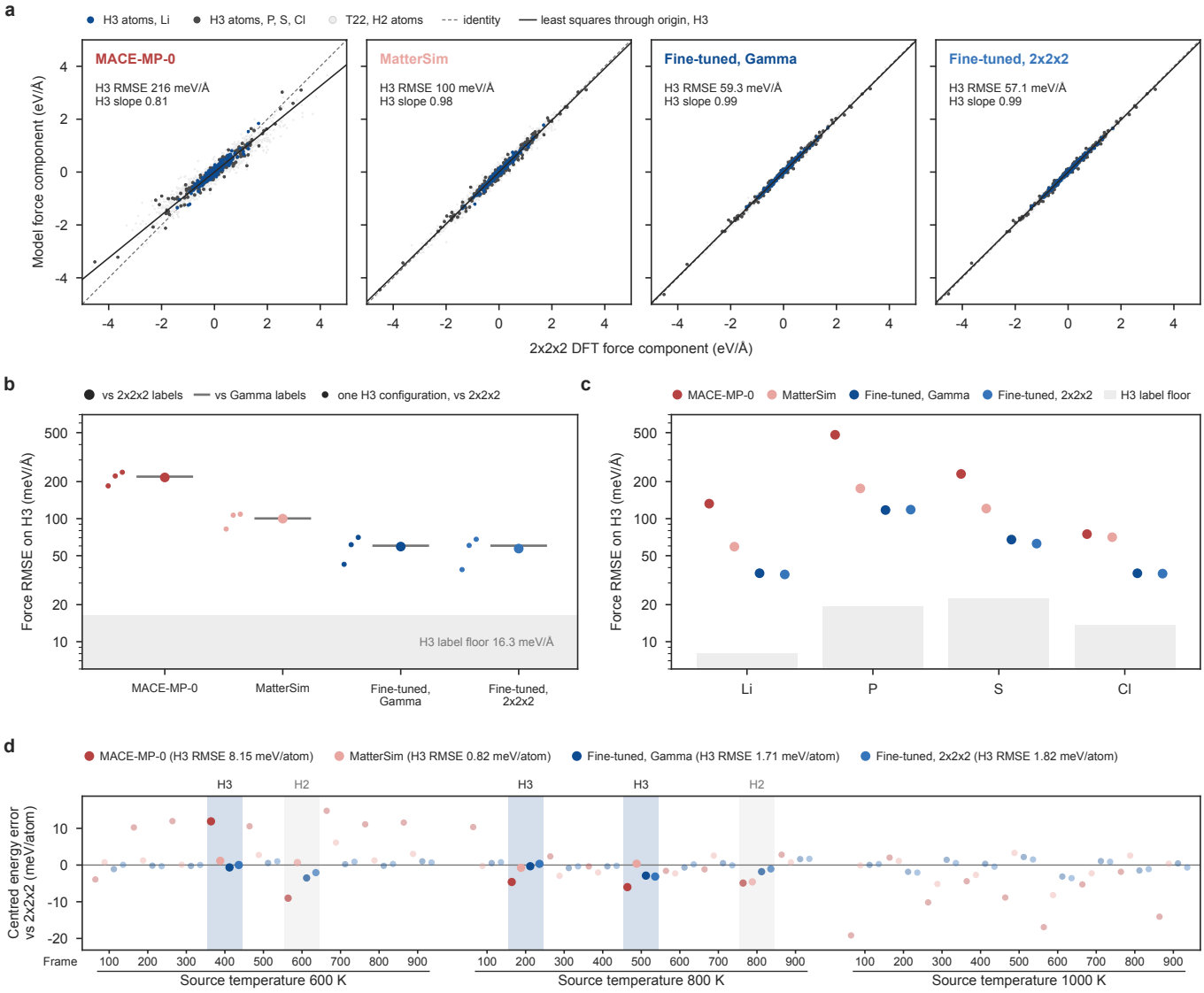

a
H3 atoms, Li
H3 atoms, P, S, Cl
T22, H2 atoms
identity
least squares through origin, H3
MACE-MP-0
H3 RMSE 216 meV/Å
H3 slope 0.81
MatterSim
H3 RMSE 100 meV/Å
H3 slope 0.98
Fine-tuned, Gamma
H3 RMSE 59.3 meV/Å
H3 slope 0.99
Fine-tuned, 2x2x2
H3 RMSE 57.1 meV/Å
H3 slope 0.99
Model force component (eV/Å)
2x2x2 DFT force component (eV/Å)
b
vs 2x2x2 labels
vs Gamma labels
one H3 configuration, vs 2x2x2
Force RMSE on H3 (meV/Å)
H3 label floor 16.3 meV/Å
MACE-MP-0
MatterSim
Fine-tuned, Gamma
Fine-tuned, 2x2x2
c
MACE-MP-0
MatterSim
Fine-tuned, Gamma
Fine-tuned, 2x2x2
H3 label floor
Force RMSE on H3 (meV/Å)
Li
P
S
Cl
d
MACE-MP-0 (H3 RMSE 8.15 meV/atom)
MatterSim (H3 RMSE 0.82 meV/atom)
Fine-tuned, Gamma (H3 RMSE 1.71 meV/atom)
Fine-tuned, 2x2x2 (H3 RMSE 1.82 meV/atom)
H3
H2
Centred energy error vs 2x2x2 (meV/atom)
Frame
Source temperature 600 K
Source temperature 800 K
Source temperature 1000 K